

Froth image feature engineering-based prediction method for concentrate ash content of coal flotation

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ABSTRACT

Machine vision and machine learning have been researched widely in froth flotation and the technology continues to benefit from advances in computer technology. The feature engineering leads to a predicted resistance in the machine learning pipelines, especially in coal flotation. This study sought to examine the performance of feature engineering of coal flotation froth image on ash content prediction with an industrial dataset. In order to evaluate the practical use in industry, the morphoscopic (3 features), statistical (gray levels histogram (5 features), gray level co-occurrence matrix (24 features), statistical modeling (48 features)) and color spaces (18 features) are used to prepare feature engineering. Correlation matrix are used to investigate the relationship between features and ash content. The support vector regression is used to predict ash content. The evaluation of the model performance shows that the principal component analysis can effectively improve the accuracy. When the feature dimension is reduced to 14 by the principal component analysis, the optimal RMSE is 0.6331, the R^2 value is 0.78. The feature engineering of coal flotation froth image in this paper can make a good prediction of the coal flotation concentrate ash content. Furthermore, the results can be used as the theoretical basis for the intelligent construction of flotation.

1. Introduction

Coal pulp flotation is an effective separation technology based on the differences in surface hydrophobicity between coal and gangue particles (Niu et al., 2018; Laskowski et al., 2001). In this process, the coal particles adhere to the bubbles and float to the flotation cells surface, which leads to the information of flotation conditions being reflected on the froth layer (Ozdemir et al., 2009; Hubbard, 2004). Unlike other minerals, the quality of coal flotation concentrate is expressed by ash content (Demir et al., 2008). The conventional method of ash content determination is based on the time-consuming processing of sampling, filtrating, drying, sample preparing and burning to ash in a muffle furnace. If flotation variables are adjusted according to this delayed result, the ash content of the flotation concentrate could not meet the production requirements. Hence, the operator still controls the flotation variables based on the froth visual features observed at the flotation cells surface. But the accuracy of this subjective control is still poor. In addition, there are some quick detection methods, such as γ -ray transmission (Yazdi and Esmaeilnia, 2008; Fookes et al., 1983), neutron analysis (Sibiya et al., 2014). But due to the high cost, harm to the

operator and the environment, there are few applications in the flotation field.

Machine vision as an efficient, environmental technology has been rapidly applied in object recognition (Serban et al., 2020), object detection (Zhang et al., 2020) and medical image analysis (Zhang et al., 2020). With the assistance of machine vision technology, many systems and pipelines have been already or expectantly applied in the industrial field in recent years (Aldrich et al., 2010; Morar et al., 2012; Nakhaei et al., 2019). Of course, the combination of machine vision and mineral flotation has been paid more attentions (Aldrich et al., 2018). Copper flotation froth image recognition has been the most widely studied. The copper concentrations in flotation were predicted based on the flotation variables containing bubble size (Saghatoleslam et al., 2004; Hosseini et al., 2015; Navia et al., 2016; Jahedsaravani et al., 2017; Cheng et al., 2020), bubble rate (Morar et al., 2006), burst rate (Morar et al., 2006), color (RGB) (Jahedsaravani et al., 2014), stability (Jahedsaravani et al., 2017; Morar et al., 2006; Barbican et al., 2007), solids loading (Morar et al., 2012; Morar et al., 2006) and velocity (Jahedsaravani et al., 2016; Geladi, 2010). The machine vision in flotation is also used for other minerals like bauxite (Aldrich et al., 2010; Gui et al., 2013; Cao et al.,

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2013), sulfide (Bonifazi et al., 2001), zinc (Kaartinen et al., 2006), gold-antimony (Wang et al., 2018; Zhou et al., 2020) and iron (Wang et al., 2014).

In coal production, the ability to predict flotation outcomes is of great importance. At present, control of flotation includes regulation of the density of the flotation feed, the reagent consumption, the air consumption, and the pulp level in the flotation tank (Petukhov V N et al., 2019). The coal products must satisfy the requirements for various applications, the ash content is the most important variable of customer requirements. In addition, accurate control of ash content can improve the recovery of clean coal, prevent material waste (Tan et al., 2016). The ash content of flotation concentrate must be strictly controlled to maximize financial profitability. Generally, the ash content in the flotation concentrate was predicted based on the analysis of the bubble number (Hargrave et al., 1996), mean bubble diameter (Citir et al., 2004; Amankwah and Aldrich, 2014), average gray (Amankwah and Aldrich, 2014), variance, energy (Zhang et al., 2014), smoothness and entropy (Massinaei et al., 2019) using BP neural network. Tan et al. (2016) investigated the relationship between the froth features (average gray level, homogeneity, bubble burst) and the concentrate ash content in the laboratory. More recently, Massinaei et al. (2019) successfully analyzed relationships between the froth features and the operating conditions as well as the ash content in an industrial flotation column. However, these features are relatively single and in a lower dimension, and are not sufficient to systematically investigate the correlation between froth image features and the concentrate ash content.

Feature engineering is the process that integrates feature generation and feature selection aiming to transform the feature dimension to obtain the best prediction effect (Yang et al., 2013). For the flotation froth images, each feature space may comprise redundant, irrelevant and relevant features, reduce feature dimensions was important for extracting useful information in feature engineering (Zhang et al. 2016; Kambhatla et al. 1997). Therefore, systematically extracting the multi-dimension features of the froth image and selecting the appropriate number of feature main components is the necessary condition for accurate prediction of coal flotation concentrate ash content.

In this paper, the relationship between the froth image features and the ash content was explored systematically based on the dataset from industrial field. The multi-dimensional features of the froth image were mainly extracted from the feature engineering of morphoscopic, statistical and color spaces. Correlation matrix are used to investigate the relationship between features and ash content. Then the principal component analysis (PCA) algorithm is used for the reduction of feature dimension. Based on that, support vector regression (SVR) was used to train the model of the feature principal components data and ash content. This work can be used to control coal flotation variables and as a basis for further study.

2. Materials and methods

2.1. Materials

In general, the hardware of machine vision consists of industrial camera, lens, light source, hood, computer, etc. (Nakhaei et al., 2019) In this study, the five megapixels color industrial camera (BALSER ace-2440-20gc) with charge-coupled device (CCD) sensors, prime lens (COMPUTER, 35 mm), diffuse bowl-shaped light-emitting diodes (LED), hood and computer (Windows 10) were used to capture froth images. The camera was fixed vertically by the bracket 50 cm above the froth layer surface in the hood. The high-quality images were obtained with an optical magnification of 0.16, and the field of view area was 15.7 cm × 16.9 cm. The froth image sequence was derived at a frame rate at 17 frames/s.

The image acquisition devices were installed above the second cell of flotation machine in a coal preparation plant in Zaozhuang, China. The 1/3 coking coal sample used for flotation was collected from Zaozhuang

Mining area in China. The froth image acquisition set-up as realized in coal flotation circuit is shown in Fig. 1. Cover the ambient light and turn on the diffuse bowl-shaped LED light source. The froth images were automatically and periodically acquired by an image acquisition software system. At the same time, the froth samples were taken from the cell overflow port directly and record the sampling time accurately.

2.2. Dataset

The datasets were obtained under normal production conditions at the coal preparation plant. The image acquisition software system was used to capture images periodically and the size of the output images is 2048 × 2048 pixels. Continuous three-frames images were captured for each ash content value-correspondingly. Fig. 1 shows the location of the sample, records the accurate sampling time, and then sends the sample to the laboratory to test ash content. The slow-ashing method is used to test the ash content (Kambhatla and Leen, 1997). Randomly sampled 15–20 times every day. After about two months, 984 ash content samples and 2952 images were collected after removing some fuzzy images.

2.3. Morphoscopic features

Woodburn et al. firstly confirmed in the laboratory that the mean bubble size is related to a low-rank UK coal content during the flotation, and the consistency of the predictions with the experimental data was encouraging (Woodburn et al., 1994). In this study, the contrast between the highlight spot of the bubble and its corner regions was extremely uneven on the froth image. The watershed segmentation algorithm with illumination modeling-based marker extraction is used to extract bubble morphoscopic features (Zhang et al., 2018). The segmentation effect is shown in Fig. 2. The size of the cumulative pixel unit was used to represent the morphoscopic features. Suppose the size of a single-pixel was 1.20 × 1.20 mm. The morphoscopic features of froth image mainly extract the bubble size mean and the bubble inscribed circle diameter means, and the number of bubbles.

2.4. Statistical features

Gray Levels Histogram. Gray levels histogram features were considered to be the easiest first-order statistical category feature of image (Prats-Montalbán et al. 2011). They characterize the image based on the statistics of the single-pixel level. In this study, the mean, median, variance, skewness and kurtosis were extracted. These features can be calculated using the following formulas:

$$Mean = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W P(i,j) \quad (1)$$

$$Median = median(P(i,j)) \quad (2)$$

$$Variance = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (P(i,j) - Mean)^2 \quad (3)$$

$$Skewness = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \left[\frac{P(i,j) - Mean}{\sqrt{Variance}} \right]^3 \quad (4)$$

$$Kurtosis = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \left[\frac{P(i,j) - Mean}{\sqrt{Variance}} \right]^4 - 3 \quad (5)$$

Where H , W represent the height and width of the froth image, respectively. $P(i,j)$ is the element (i,j) of gray matrix of froth image.

Gray level co-occurrence matrix. The Gray Level Co-occurrence Matrix (GLCM) was first reported by Haralick (1979). The GLCM was defined as the probability that the range of the pixel from gray level n to m with an angle of θ and step of d . The GLCM of the froth image is shown in Fig. 3.

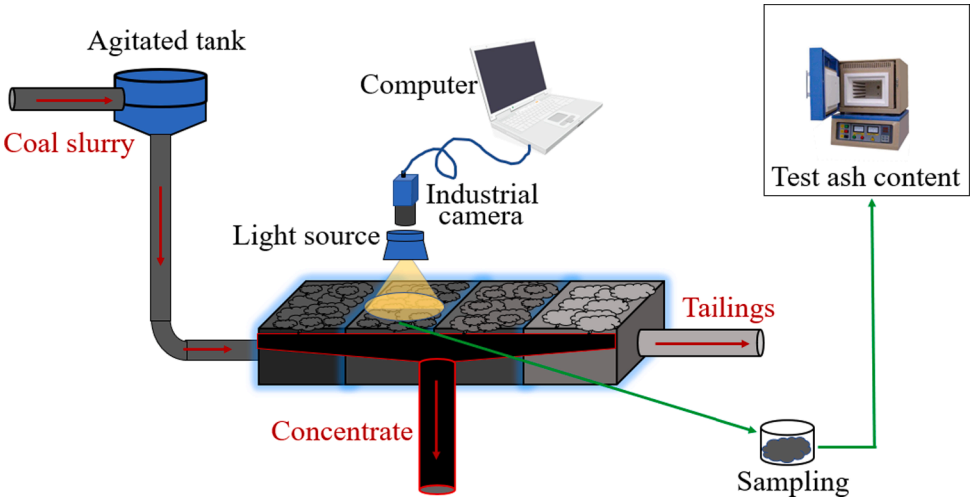


Fig. 1. Acquisition set-up as realized in coal flotation circuit.

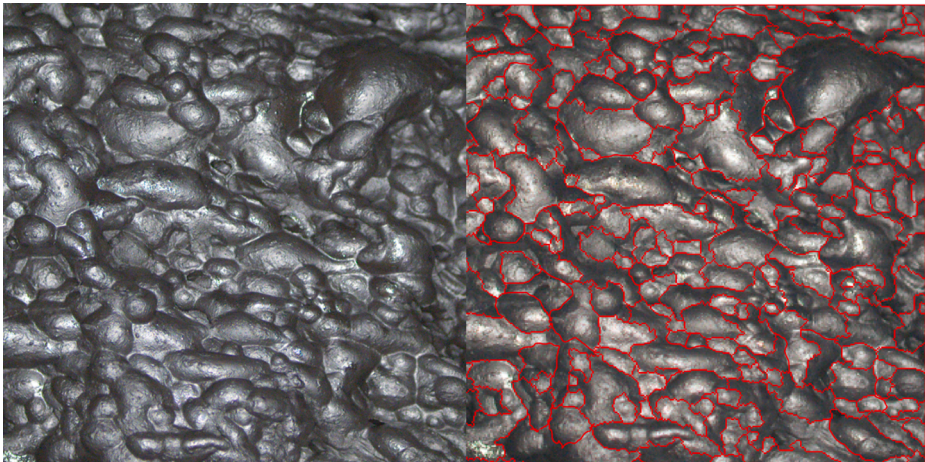


Fig. 2. Segmentation of coal flotation froth image.

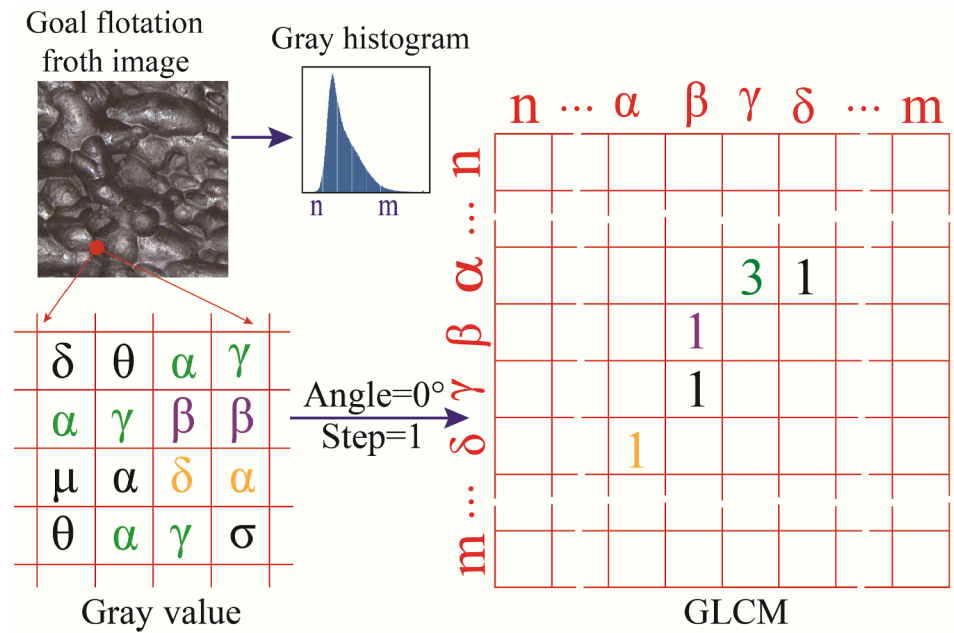


Fig. 3. Create a GLCM (Angle = 0°, Step = 1).

The size of the GLCM matrix was driven by the number of grey-levels ($m+1$) defining the image. In this paper, the most relevant texture features extracted from GLCM were contrast, homogeneity, energy, correlation, angular second moment (ASM) and entropy. These features can be calculated as follows:

$$\text{Contrast} = \sum_{i,j=0}^{N-1} P(i,j)(i-j)^2 \quad (6)$$

$$\text{Homogeneity} = \sum_{i,j=0}^{N-1} \frac{P(i,j)}{1 + (i-j)^2} \quad (7)$$

$$\text{Energy} = \sum_{i,j=0}^{N-1} \sqrt{P(i,j)^2} \quad (8)$$

$$\text{Correlation} = \sum_{i,j=0}^{N-1} \frac{(ij)P(i,j) - (\mu_x \times \mu_y)}{\sigma_x \times \sigma_y} \quad (9)$$

$$\text{ASM} = \sum_{i,j=0}^{N-1} P(i,j)^2 \quad (10)$$

$$\text{Entropy} = \sum_{i,j=0}^{N-1} P(i,j)(-\ln P(i,j)) \quad (11)$$

Where N is the number of the component levels in the image under quantization, $P(i, j)$ is the element (i, j) of GLCM.

Statistical modeling. Gabor wavelet transform has been recognized as a very useful tool in texture analysis due to its optimal localization properties in both spatial and frequency domains was similar to the human visual system (Arrospide and Salgado, 2013). The Gabor feature map is shown in equation (12), which was generally obtained by convolving the image to be processed with the Gabor kernel.

$$F_{u,v}(i, j) = P(i, j) \otimes G_{u,v}(i, j) \quad (12)$$

Where $P(i, j)$ is the gray value of the image at (i, j) , $G_{u,v}(i, j)$ is the Gabor kernel function, whose scale is u and direction is v , $F_{u,v}(i, j)$ is the results of Gabor filter.

In general, Gabor wavelet is expressed mainly by the real and imaginary part. Their relationship is shown as follows:

$$M_{u,v}(i, j) = \sqrt{[Re(F_{u,v}(i, j))]^2 + [Im(F_{u,v}(i, j))]^2} \quad (13)$$

Where $Re(F_{u,v}(i, j))$ and $Im(F_{u,v}(i, j))$ represent the real and imaginary part, respectively. $M_{u,v}(i, j)$ is Gabor amplitude feature.

According to the complexity of coal flotation pulp, it was necessary to let neighbor pixels participate in the composition of Gabor amplitude features for the extraction of generalized features. The specific approach was to divide the froth image into N subplots with boundaries as m pixels, and then calculate the mean of all subplot amplitude features. The new magnitude was mean weighted by the mean amplitude and the amplitude of the pixel, which calculated as follows:

$$M^{(s)}_{u,v}(i, j) = \frac{\left[\sum_{x,y=1}^m M_{u,v}(x, y) \right] / m + M_{u,v}(i, j)}{2} \quad (14)$$

$$M^{(new)}_{u,v,N} = \frac{\sum_{z=0}^{N-1} M^{(s)}_{u,v}(i, j)}{m^2 / N} \quad (15)$$

Where $M^{(s)}$ is the amplitude of a subplot, $M^{(new)}$ is new amplitude of froth image, m is pixel length of the subplot, N is the number of subplots.

Previous studies have shown that amplitude is more in line with the gamma distribution (Γ D) (Guo et al., 2014). Hence, the probability density function (PDF) can be used to estimate model feature variables

of the Gabor wavelet amplitude feature by using maximum likelihood estimation (MLE). The calculation of the variables is as follows:

$$f(x; \alpha, \beta) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \exp^{-x/\beta} \quad (16)$$

$$\Gamma(\alpha) = \int_0^\infty \exp^{-t} t^{\alpha-1} dt \quad (17)$$

Where f is the formula for the PDF, $\Gamma(\alpha)$ is the gamma function, x is the resulting amplitude feature of Gabor wavelet, α, β are the gamma distribution variables.

Thus, the statistical modeling features of froth image can be represented as:

$$F(\text{statistical modeling}) = \{(\alpha, \beta) | u = 0, \dots, U, v = 0, \dots, V\} \quad (18)$$

Where α, β is the gamma distribution variables, U, V is the scale and directions of Gabor wavelet filters, respectively.

2.5. Color spaces features

H (Hue), S (Saturation) and I (intensity) were similar to the humans perceive so that they are important for froth image recognition. the conversion formula from RGB color space to HIS color space is shown in Equation 19. A color co-occurrence matrix (CCM) method was proposed to measures the spatial color distribution features of froth image in color space (Palm et al., 2004). The calculation of the CCM is shown in Fig. 4. In this study, the HS, HI, SI three CCM were used to transmit the features. The six generalize-used features including Contrast, Dissimilarity, Homogeneity, Energy, Correlation, Angular Second Moment, Entropy are calculated the same as formula 6–11 in section 2.4 (Gray level co-occurrence matrix).

$$\begin{cases} I = \frac{1}{3}(R + G + B) \\ S = 1 - \frac{3 \times [\min(R, G, B)]}{R + G + B} \\ H = \begin{cases} \theta & G \geq B \\ 2\pi - \theta & G < B \end{cases} \end{cases} \quad (19)$$

$$\text{where } \theta = \arccos \left\{ \frac{[(R - G) + (R - B)]/2}{[(R - G)^2 + (R - B) \cdot (G - B)]^{1/2}} \right\}$$

2.6. Methodology used to analysis the features

The Pearson's correlation matrix is generated from features of froth image and ash content, which can be used to describe the correlations among N feature variables and ash content. The correlation matrix is a square symmetrical $N \times N$ matrix and calculated by the Python code according to the method of Ref (Pearson, 1900). The correlation factors are between -1 and 1 (0 : no correlation and ± 1 : high correlation). A positive value (or negative value) indicates that the two correlated variables increase (or decrease) simultaneously. In this paper, the correlation between variables is described in the significance level case of $0.5, 0.01$ and 0.001 , respectively.

In addition, the PCA is one of the effective means of data reduction and dimension reduction (Kambhatla et al. 1997). For an n -dimension feature data $X = \{X_1, X_2, \dots, X_n\}$. The process of extracting m principal components using PCA is as follows:

First normalize the original matrix X and get matrix $Y = \{Y_1, Y_2, \dots, Y_n\}$. The covariance matrix Z is calculated according to the formula (20):

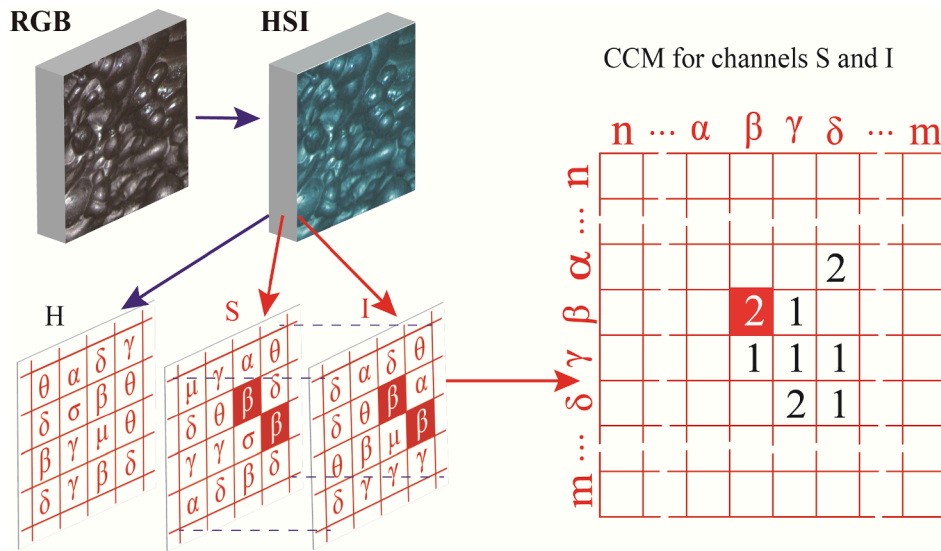


Fig. 4. Create a CCM for channels S and I.

$$z_{ij} = \frac{\sum_{k=1}^n (y_{ki} - \bar{y}_i)(y_{kj} - \bar{y}_j)}{\sqrt{\sum_{k=1}^n (y_{ki} - \bar{y}_i)^2 \sum_{k=1}^n (y_{kj} - \bar{y}_j)^2}} \quad (20)$$

Where z_{ij} is the element in coefficient matrix Z , y_{kj} is the element in normalized matrix Y , n is the dimension of the features.

The eigenvalue and eigenvector of the coefficient matrix Z were calculated according to equation $Zu = \lambda u$. Such as $\{(\lambda_1, u_1), (\lambda_2, u_2), \dots, (\lambda_n, u_n)\}$. The cumulative contribution rate K was used to extract the principal components. In general, the cumulative contribution rate for the principal components was $>85\%$. K is calculated as follow:

$$K = \sum_{i=1}^m \lambda_i / \sum_{k=1}^n \lambda_k \quad (21)$$

Where m is the total number of principal components, n is the dimension of the features, λ is the eigenvalue.

Finally, the m principal components of the original feature data X are obtained by formula (22):

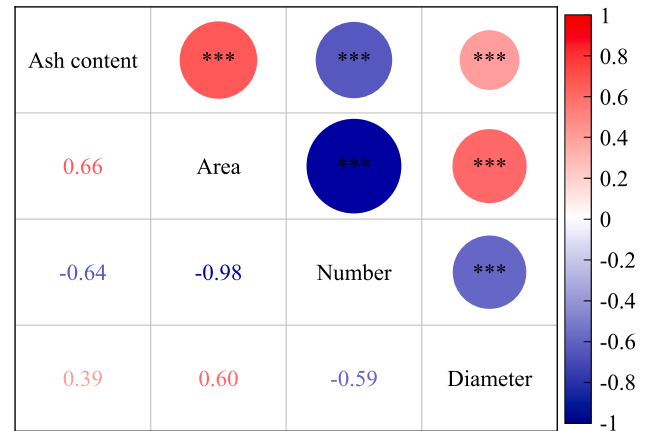
$$P_k = u_k X_1 + u_k X_2 + \dots + u_k X_n \quad (k = 1, 2, \dots, m) \quad (22)$$

Where u_k is the orthogonalization unit eigenvector corresponding to the eigenvalue λ_k .

3. Results and discussion

3.1. Morphoscopic features generation

The morphoscopic features were important variables describing the flotation performance. After the froth image was segmented by the watershed algorithm, the morphological features of the bubbles were calculated in pixels. The correlation results between morphoscopic features and ash content are shown in Fig. 5. The area (mean bubble area) and diameter (mean circumscribed circle diameter) increase with the ash content as a whole, and the correlation between area and ash content is strongest (correlation factors of 0.66). This area change was consistent with the previous research results in the laboratory (Citir et al., 2004; Tan et al., 2016). Furthermore, the study found that the mean bubble area and ash content tend to be a non-linear relationship, and the bubble number was negatively related to the ash content. Compared with the area results, the correlation between the number of bubbles and the ash content was reduced (correlation factors of -0.64). This indicates that the bubbles may be unevenly distributed during



* $p \leq 0.05$ ** $p \leq 0.01$ *** $p \leq 0.001$

Fig. 5. Correlation matrix for morphoscopic features and ash content.

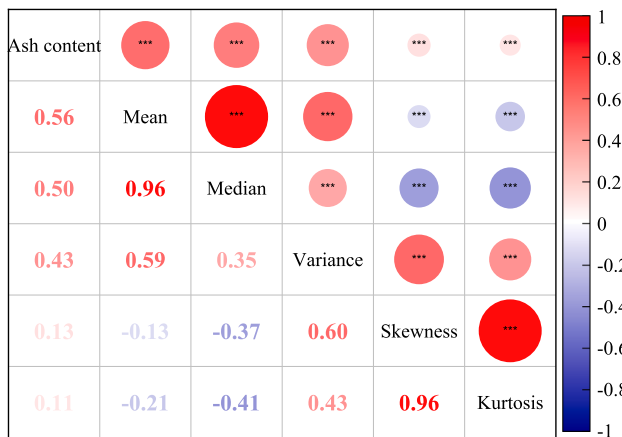
flotation, which may be caused by bubbles merge or inaccurate image segmentation.

For the three morphological features, the correlation between area and number was strongest (correlation factors of -0.98), shown a negative correlation. On the contrary, the positive correlation between area and diameter indicates that the shape of the flotation foam is mostly a rule shape close to the circle.

3.2. Statistical features generation

3.2.1. Gray levels histogram

The gray value of the froth image is the earliest feature used to predict coal flotation performance (Wang et al., 2014; Tan et al., 2016). In this study, the gray levels histogram distribution of the froth image was stable and mostly obeyed gamma distribution. Based on the industrial field dataset in this study, the mean, variance, median, skewness and kurtosis were extracted in Fig. 6. The value of gray levels histogram features increases with the ash content. The results of the correlation matrix show that there is a high correlation between the mean, variance, median and ash content, of which the mean is strongest (correlation factors of 0.56). It is due to the transition of the gray level of coal and gangue. The results of skewness and kurtosis indicate that they have



* $p \leq 0.05$ ** $p \leq 0.01$ *** $p \leq 0.001$

Fig. 6. Correlation matrix for gray levels histogram features and ash content.

little to do with ash content. The lowest correlation factor of 0.11 was obtained by kurtosis. For five gray levels histogram features, the correlation of the mean – median and the skewness – kurtosis were high (correlation factors of 0.96). But they have few to do with ash content.

3.2.2. Gray level co-occurrence matrix

The froth image has 256 distinct grey levels, as the results of the size of the GLCM were statistically calculated to 256×256 . Sets two variables that described the pixel pairs: pixel pairs spacing (1, 3, 5 pixels) and pixel pairs direction (0° , 45° , 90° , 135°). The study found that the GLCM features with pixel pair spacing of 3 were better fitted with ash content. The correlation between them was organized into a correlation matrix as shown in Fig. 7. The contrast, homogeneity and correlation were positively related to ash content, the ash content increases with these values. The correlation in the horizontal direction (0°) has the highest correlation with ash content (correlation factors of 0.52).

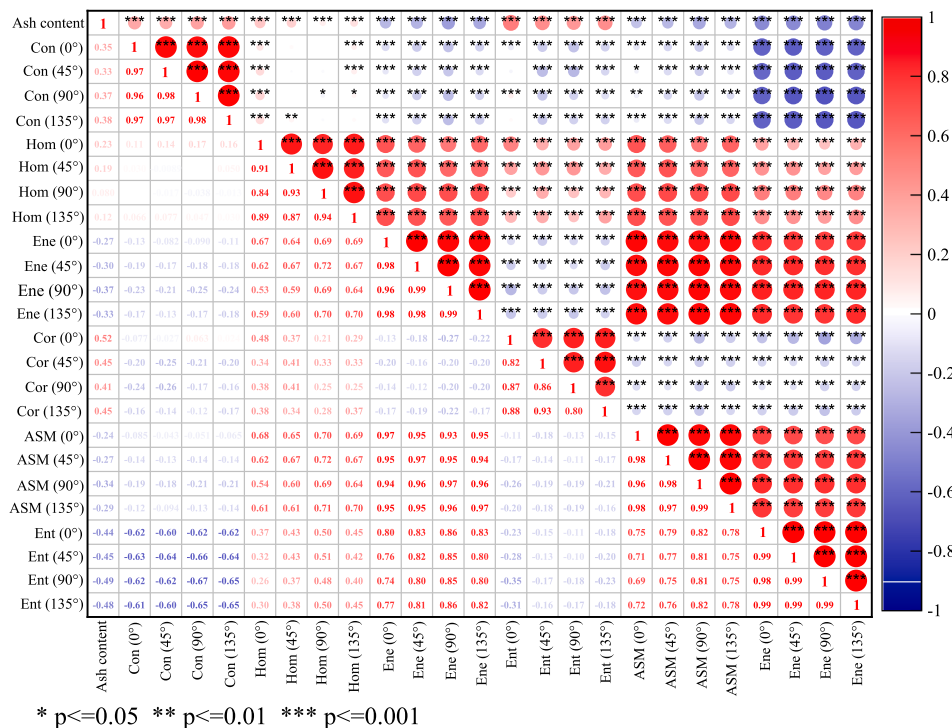
Conversely, energy, ASM and entropy were inversely proportioned to ash content. Entropy has the strongest negative correlation with ash content, and the correlation was a little different from each angle (correlation factors of -0.49). This result was slightly different from Tan et al. (2016). It may be that the industrial flotation froth size span of single-cell was not as large as the laboratory. For flotation froth, as the ash content increases, the bubble size expands to make the texture of the boundary more obvious (contrast increases). The increase in correlation indicates that the gray level values changed between local pixels was smoother. Meanwhile, the decrease of entropy indicates that the changes in the size of bubbles also lead to changes in the reflective region of its surface.

For GLCM features, the correlation between contrast and correlation was low. Some values were not significant at $P > 0.5$, and the two features themselves are equally less relevant. Moreover, there is a high correlation between features such as homogeneity, energy, ASM and entropy, and the correlation between Energy and ASM was best (correlation factors of 0.97).

3.2.3. Statistical modeling

Gabor filter generates a complete but non-orthogonal basis set to describe the local frequency of the froth image (Chen et al., 2011). In this study, the 24 sub-bands feature images were generated by a Gabor filter in six scales (2, 3, 4, 5, 6, 7 pixels) and four orientations (0° , 45° , 90° , 135°). Fig. 8 illustrates the frequency response of the Gabor filter bank and the amplitude distribution of one of the sub-bands (scale 3, angle 0°). It can be seen that the distribution of the sub-bands was more fitted to the gamma distribution. This result was consistent with the result of Guo et al. (2014) on the vehicle verification task. Hence, the gamma distribution model variables (α and β) can be estimated by the PDF of gamma distribution by using maximum likelihood estimation (MLE).

The Gabor filter discards detailed information about the local bubble structure. The froth image was partitioned into sub-bands, and each sub-band was processed as a separate froth image. Figure S1 shows the correlation between the statistical results of the gamma distribution



* $p \leq 0.05$ ** $p \leq 0.01$ *** $p \leq 0.001$

Fig. 7. Correlation matrix for GLCM features and ash content (Con: contrast, Hom: homogeneity, Ene: energy, Cor: correlation, ASM: angular second moment, Ent: entropy).

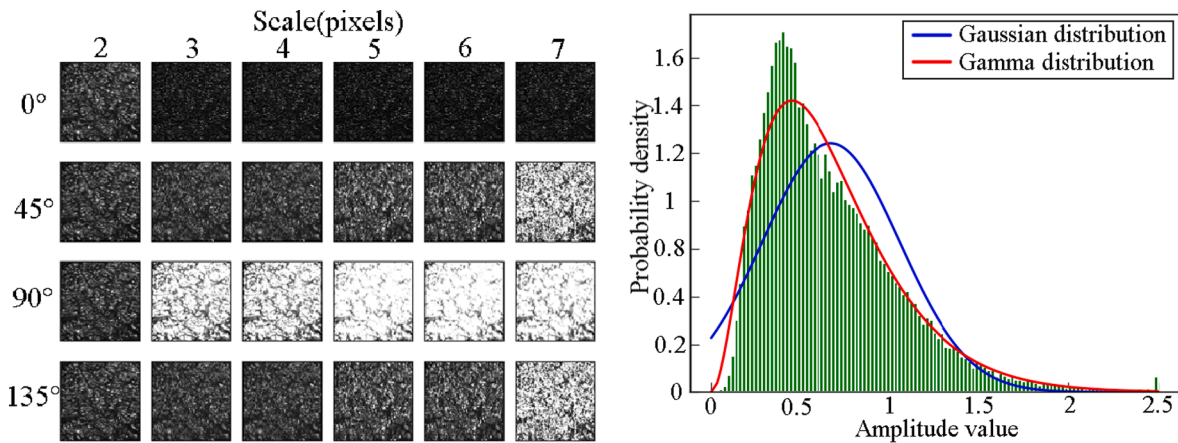


Fig. 8. The Gabor filter bank and an amplitude distribution (scale 3, angle 0°) of the sub-bands.

model variables and the ash content. Most features of the filtering direction of 45° and 90° were positively related to ash content, the ash content increases with these values. The correlation between ash content and features in other directions was poor, and some features in the direction of 135° were not significant at $P > 0.5$. The highest correlation coefficient was obtained by α (90°, 3p) and α (90°, 4p) (correlation factors of 0.36). On the other hand, the features of the 90° filter direction were poorly related to other features, but they are highly self-related, and other features are also highly self-related (correlation factors were basically >0.8).

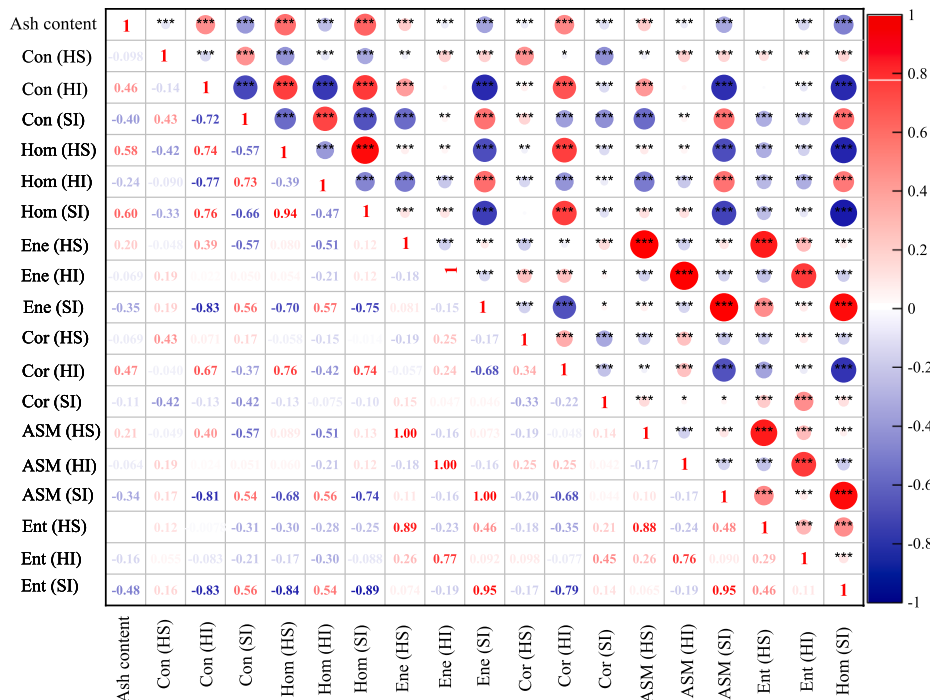
3.3. Color spaces features generation

Color co-occurrence matrix (CCM) uses HSI color features instead of the visual of the flotation driver to evaluate the coal flotation conditions (Gui et al., 2013; Ma et al., 2012). Similar to GLCM, features were extracted from three CCMs (HS, HI, SI). The correlation between CCM

features and ash content is shown in Fig. 9. It can be seen from the correlation matrix that there is a correlation between CCM-based texture features and ash content in addition to Ent (HS) (not significant at $P > 0.5$), of which Hom (SI) has the highest correlation with ash content (correlation factors of 0.60). Therefore, there was no much texture information stored in the H channel. The features associated with the ash content were mainly reflected in the S and I channel, this may be due to the froth image in color space was single. To sum up, correlation (HI), entropy (SI) and homogeneity (HS, SI) were better than used other features to evaluate coal flotation.

3.4. SVR model training

In the above section, the corresponding features of morphoscopic, statistical and color spaces were extracted to estimate the ash content of coal flotation concentrate. As shown in Fig. 10, all 99 image features were expressed as a correlation matrix. It can be seen that the gray levels



* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

Fig. 9. Correlation between CCM features and ash content (Con: contrast, Hom: homogeneity, Ene: energy, Cor: correlation, ASM: angular second moment, Ent: entropy).

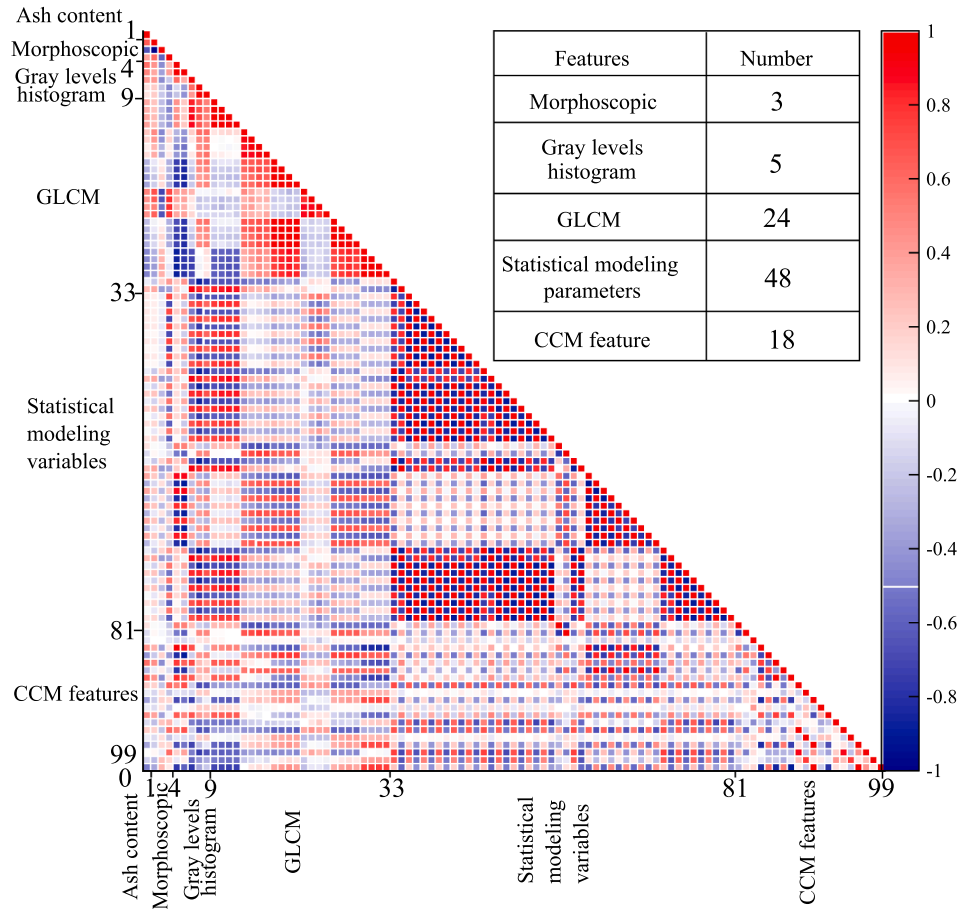


Fig. 10. Correlation matrix for froth image feature engineering.

histogram features were relatively highly relevant to other features in feature engineering. Table 1 lists the positions of all features in the correlation matrix (Fig. 10).

Table 1

The number of all features in the feature engineering (number is the coordinate in Fig. 11.) (Con: contrast, Hom: homogeneity, Ene: energy, Cor: correlation, ASM: angular second moment, Ent: entropy).

Type	Number	Features
Morphoscopic	2–4	bubble area; number of bubbles; circumscribed circle diameter.
Gray levels histogram	5–9	Mean; variance; median; skewness; kurtosis.
GLCM	10–33	Con (0°); Con (45°); Con (90°); Con (135°); Hom (0°); Hom (45°); Hom (90°); Hom (135°); Ene (0°); Ene (45°); Ene (90°); Ene (135°); Cor (0°); Cor (45°); Cor (90°); Cor (135°); ASM (0°); ASM (45°); ASM (90°); ASM (135°); Ent (0°); Ent (45°); Ent (90°); Ent (135°).
Statistical modeling variables	34–81	$\alpha(0^\circ, 2p)$; $\beta(0^\circ, 2p)$; $\alpha(0^\circ, 3p)$; $\beta(0^\circ, 3p)$; $\alpha(0^\circ, 4p)$; $\beta(0^\circ, 4p)$; $\alpha(0^\circ, 5p)$; $\beta(0^\circ, 5p)$; $\alpha(0^\circ, 6p)$; $\beta(0^\circ, 6p)$; $\alpha(0^\circ, 7p)$; $\beta(0^\circ, 7p)$; $\alpha(45^\circ, 2p)$; $\beta(45^\circ, 2p)$; $\alpha(45^\circ, 3p)$; $\beta(45^\circ, 3p)$; $\alpha(45^\circ, 4p)$; $\beta(45^\circ, 4p)$; $\alpha(45^\circ, 5p)$; $\beta(45^\circ, 5p)$; $\alpha(45^\circ, 6p)$; $\beta(45^\circ, 6p)$; $\alpha(45^\circ, 7p)$; $\beta(45^\circ, 7p)$; $\alpha(90^\circ, 2p)$; $\beta(90^\circ, 2p)$; $\alpha(90^\circ, 3p)$; $\beta(90^\circ, 3p)$; $\alpha(90^\circ, 4p)$; $\beta(90^\circ, 4p)$; $\alpha(135^\circ, 3p)$; $\beta(135^\circ, 3p)$; $\alpha(135^\circ, 4p)$; $\beta(135^\circ, 4p)$; $\alpha(135^\circ, 5p)$; $\beta(135^\circ, 5p)$; $\alpha(135^\circ, 6p)$; $\beta(135^\circ, 6p)$; $\alpha(135^\circ, 7p)$; $\beta(135^\circ, 7p)$.
CCM	82–99	Con (HS); Con (HI); Con (SI); Hom (HS); Hom (HI); Hom (SI); Ene (HS); Ene (HI); Ene (SI); Cor (HS); Cor (HI); Cor (SI); ASM (HS); ASM (HI); ASM (SI); Ent (HS); Ent (HI); Ent (SI).

The dataset in this study was collected over a two-month period at the industrial field and consisted of 2952 froth images and 984 ash content samples. Each ash content label corresponds to a continuous three-frame image. Train and test constitute 90% and 10% of the dataset and were generated randomly. Moreover, the variables of C (penalty parameter), ϵ (Insensitive loss parameter), and γ (kernel function parameter) were important for SVR model training. Grid search method was used to optimize models and search for hyperparameters (Laurier et al., 2007; Sun and Huang, 2006). During model training, 10-fold cross-validation was used to divide the input dataset and test the accuracy of the model (Lo et al., 2015). The step of cross-validation was to divide the training set into nine separate subsets, one of which was left for testing and the rest are used for training a model. Each subset was repeated 10 times and the mean of the 10-fold-related error values was calculated as a cross-validation error. Grid search finds the best combination of hyperparameters with the lowest error.

In order to evaluate the performance of the SVR models, root mean square error (RMSE) of the predictions was used to calculate the error in this paper. It is calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (18)$$

Where y_i and $\bar{y}_i$ are the actual and the predicted ash content, n is the number of samples in the test set.

In addition, PCA has arguably been the most widely used multivariate data analysis tool for dimensionality reduction (Nakhaei et al., 2019). Train SVR models, the model results using all 98 features and features determined by PCA (the principal component) are given in Table 2. The model was coded using Python 3.7 and a third-party library

Table 2

Prediction results.

Number of principal components	Proportion of the original information (%)	RMSE	R ²
2 (PCA)	45.96	1.6543	0.49
4 (PCA)	67.67	1.1684	0.52
6 (PCA)	84.53	0.7876	0.55
8 (PCA)	89.22	0.6921	0.63
10 (PCA)	93.57	0.6639	0.69
12 (PCA)	95.14	0.6426	0.74
14 (PCA)	97.32	0.6331	0.77
16 (PCA)	98.16	0.6327	0.78
All features	100	0.9354	0.61

(scikit-learn 0.20.3, numpy 1.18.1). It can be seen from Table 2 that the PCA dimensionality reduction method improves the predictive performance of the model. After all features are input into the SVR model for training, the RMSE is 0.9354. When the features are reduced to 14 principal components, the optimal RMSE is 0.6331. For test set, the determination coefficient (R^2) is used to represent the prediction performance of the SVR model. As can be seen in Table 2, the R^2 value also obtained the best 0.78 at optimal RMSE, the performance of the model is best. More efficient models were obtained in lower-dimensional feature space, and the training time is reduced. Therefore, the features from feature engineering can be considered as the best factors for ash content prediction..

3.5. Predict performance

To demonstrate the effectiveness of this feature engineering for ash content prediction, the 30 froth images from the coal preparation plant (Zaozhuang, China) were used to forecast the ash content of coal flotation concentrate. The froth image case of different ash content in the validation is shown in Figure S2. The feature engineering of the froth image is established, and the features after the dimensionality reduction are entered into the SVR model. The correlation between the real ash content and the prediction is shown in Fig. 11. Most points are near the fitting line. The correlation coefficient for the validation is 0.72, and the trend of predicted values is close to the real value. It is obvious that the feature engineering has better performance and can predict the coal flotation concentrate ash content accurately and satisfactorily.

Compared to the predictions of other coal flotation variables (e.g. recovery of flotation concentrate), the R of the ash content prediction is slightly lower (E Jorjani et al., 2009). The prediction of ash content is a more challenging study in flotation monitor. In addition, the laboratory correlation analysis of ash content is significantly better than industry (Tan et al., 2016). This may be due to more complex conditions on the flotation site.

4. Conclusions

The focus of this study was to investigate the feature engineering of the froth image for the accurate prediction of coal flotation concentrate ash content. From the perspective of image processing, the features of morphoscopic (3 features), statistical (gray levels histogram (5 features), gray level co-occurrence matrix (24 features), statistical modeling (48 features)) and color spaces (18 features) were studied systematically.

The correlation matrix shows that morphology has the strongest correlation ($R = 0.66$) with gray in all features. For statistical modeling features, there is significant self-correlation, but the correlation with ash content is not high.

The PCA can effectively improve the accuracy of the SVR model. When the feature dimension is reduced to 14 by the PCA, the optimal RMSE is 0.6331. Relatively, the RMSE for direct training of all features is 0.9354.

The performance results show that the feature engineering has better

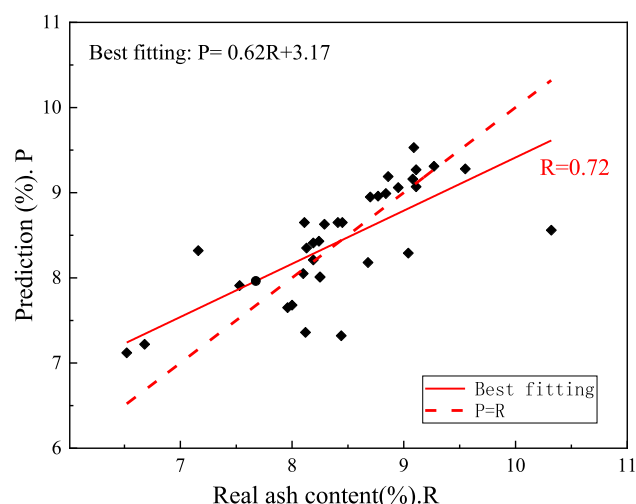


Fig. 11. The predictive performance of SVR model for ash content.

performance and can predict the coal flotation concentrate ash content accurately and satisfactorily. But, compared to the other coal flotation variables and laboratory results, the prediction of ash content is more challenging in the industry.

The used method and results can further be used to estimate the ash content of coal flotation concentrate and control the flotation process. The research results can greatly contribute to the intelligent construction of a coal preparation plant.

CRediT authorship contribution statement

Zhiping Wen: Conceptualization, Methodology, Writing - original draft. **Changchun Zhou:** Supervision. **Jinhe Pan:** . **Tiancheng Nie:** . **Ruibo Jia:** . **Fan Yang:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mineng.2021.107023>.

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